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INITIAL CONCEPT PROPOSAL | AI BUILDER CUP 2026

MadadDesk

From scattered distress messages to a live emergency map.

An AI-assisted emergency platform combining SOS reporting, a verified live map, coordinator workflows and location history. It starts with crowded commercial buildings and a focused, human-reviewed prototype.

Proposed theme	Sustainability & Social Impact
Primary users	Building emergency coordinators and authorized response partners
Initial scope	One building, text SOS reports, verified map and coordinator dashboard
Prepared for	Maryam Saba and prospective team members
Date / status	10 October 2026 / concept proposal; not a deployed emergency service

The idea in one sentence

When multiple people report a building emergency in Urdu, Roman Urdu or English, MadadDesk turns those reports into source-linked incident cards and map markers that a coordinator can verify, assign and update.

Inspired by the Gul Plaza fire in Karachi. This proposal focuses on information coordination; it does not claim that an app could have prevented that tragedy.

01 / PROBLEM

Why this problem matters

The real-world motivation: Gul Plaza

The Gul Plaza fire began on 17 January 2026 in Karachi. AP reported a multistory market with approximately 1,200 shops, rapid spread through merchandise, structural collapse and families searching for missing relatives. Reuters reported that thick smoke and limited ventilation hindered access to trapped people. [1, 2]

These reports illustrate the complexity of a large building emergency. They do not establish that fragmented reporting caused the losses. The coordination gap below is a product hypothesis to validate through interviews and drills.

The specific problem we propose to solve

During an emergency, a coordinator may receive overlapping calls, messages and second-hand reports. A message can describe the building but omit the floor or shop number. A later update can change the situation without reaching everyone. Manually reconciling these reports can make it difficult to maintain a reliable incident picture.

Information gap	Operational consequence
Unclear indoor location	A building-level GPS pin does not identify a floor, shop or landmark.
Incomplete or multilingual reports	The coordinator must interpret messages and ask for missing details.
Repeated or conflicting reports	Separate groups may be confused with duplicates, or old information may be treated as current.
No shared status history	Teams may struggle to see who owns a task and whether its status has been verified.

Who benefits first?

The initial user is a designated building coordinator managing a drill or simulated incident. Occupants submit reports; authorized responders view verified tasks. Shopkeepers and families benefit from clearer, controlled communication. Agency integration would require agreements and operational testing.

02 / SOLUTION

How MadadDesk would work

Step	Proposed behavior
1. Receive	An occupant or coordinator submits a short text report in Urdu, Roman Urdu or English. The system records the original text, time and a reference ID.
2. Structure	Gemini extracts only stated details: incident type, building, floor, shop/landmark, reported people and requested help. Missing values remain unknown.
3. Clarify	The app identifies missing details and drafts a short clarification question. Replies enrich the same report rather than creating disconnected records.
4. Compare	Possible duplicate reports and contradictions are flagged for review. The coordinator decides whether reports belong together.
5. Review and assign	The coordinator checks sources, corrects fields and assigns a task to an authorized team. The AI does not autonomously dispatch responders.
6. Track	A timeline records acknowledgement, assignment and verified updates. Stale reports are visibly marked; closure requires human confirmation.

Illustrative demo - fictional, not a Gul Plaza reconstruction

Input: "Plaza ki second floor, shop 24 ke paas dhuaan hai. Hum 3 log hain."

Draft card: Incident: smoke reported. Floor: second. Landmark: shop 24. Reported group: 3. Building: unknown. Verification: pending. Source: original message and timestamp.

Clarification: "Which plaza are you in?" After a reply, the coordinator reviews the card and assigns it. A second report from the same area is flagged as possibly related; people counts are not automatically added or merged.

What the AI contributes

Multilingual extraction, concise summaries, missing-field detection and suggested report relationships. The dashboard must remain usable through manual entry if AI processing fails. All generated fields stay connected to their source reports.

03 / DIFFERENCE

Why this approach is different

The distinction is the workflow: building-specific report understanding, verified live mapping and human-reviewed coordination. The table compares general approaches, not audited commercial products; a competitor review is still needed.

Common approach	MadadDesk proposed focus
SOS button or location sharing	Adds floor, shop and landmark details, uncertainty labels and a coordinator review step.
Group chat or message inbox	Converts scattered messages into incident records with source links, ownership and status history.
Generic AI chatbot	Supports a bounded incident workflow rather than open-ended emergency advice.
Form-based incident tracker	Accepts natural-language reports and drafts missing-information questions before human approval.

Why use AI instead of just a form?

People under stress may send short, incomplete messages in mixed languages. AI can reduce the effort of converting these messages into consistent fields. Structured fields, validation and human review still do the operational work; AI is useful only if it improves accuracy or review time in tests.

Trust and operational boundaries

No invented facts: Unknown location, people count and incident details remain unknown. Every card shows the original report. A report describes a claim, not a verified condition.

Human authority: No automated rescue decisions, evacuation route generation or police/ambulance calls. Urgency signals are review prompts, not authoritative triage.

Controlled access: Reporter contact and exact location are restricted to authorized users. Public views exclude identifying details. Define retention and deletion policies before a pilot.

Connectivity limits: No claim of working without power or network access. Show pending/failed submission states clearly; a prototype is not a replacement for established emergency channels.

04 / MAP & BACKEND

Live SOS map and long-term incident memory

The expanded product vision

A proper backend powers SOS intake, verification, live map updates and historical records. The live view answers: where is an incident currently reported or verified? The history view answers: where have incidents happened before? These are separate questions and must remain separate on the map.

Map layer	Meaning and proposed behavior
Reported / unverified	A new SOS appears in the coordinator view with its timestamp and source. It is not broadcast as a confirmed emergency.
Verified / active	An authorized coordinator confirms the incident. The public map shows a prominent marker, verified time and approved updates.
Response underway	Authorized teams see the incident location, floor/landmark, assigned task and confirmed access information when available.
Resolved / history	The active marker is cleared after human confirmation. The record remains in a separately selectable historical layer with incident date and resolution status.
Repeated incidents	A later analytics layer highlights locations with multiple recorded incidents over a chosen period. Counts are not a certified building safety rating.

Nearby alerts and movement decisions - later phase

Opt-in nearby alerts can notify users of verified active incidents and relay official instructions. Affected zones need authorized confirmation; radius alerts alone are approximate. The absence of a marker does not establish safety. Historical incidents do not prove that a building is currently unsafe, and the system does not generate a safe evacuation route from map pins.

Backend responsibilities

Store report IDs, coordinates and location accuracy, building/floor details, source text, timestamps and verification state. Keep separate incident, assignment, status-event and notification records. Enforce permissions on the server; use retries without duplicate reports, real-time subscriptions and an auditable status history. Exact reporter location and contact details stay restricted.

Public map markers identify the incident area, not individuals. Allow location correction and distinguish device GPS from a user-selected location. A report can be saved as pending, but must never appear successfully delivered before backend acknowledgement.

05 / COORDINATION

Nearby agencies, acknowledgement and mutual aid

The goal: reach responders, not only nearby occupants

An SOS should enter the incident workflow and notify relevant enrolled response agencies near the incident, rather than alerting only people inside the affected space. Candidate agencies are selected using incident type, service coverage, location, capability and reported availability. Nearby does not automatically mean suitable.

Proposed response workflow

Stage	Backend and coordinator behavior
Report and review	Record the SOS and immediately surface it to the coordinator. Flag it as unverified until reviewed; the prototype does not silently treat every report as confirmed.
Select agencies	Match the incident to enrolled agencies by coverage and capability. A fire may need fire services plus ambulance and police support.
Notify and acknowledge	Create an inbox alert and attempt push delivery. Each agency can acknowledge, accept, decline or mark itself unable to respond. Notification delivery is not acceptance.
Assign and synchronize	A coordinator confirms a lead agency and any support agencies. Atomic assignment updates prevent conflicting claims; all authorized agencies see the current ownership.
Escalate and request help	If an agency is busy or no acknowledgement arrives within a configured window, escalate to the coordinator and additional eligible agencies. Confirmed capacity gaps can trigger a mutual-aid request.
Close with evidence	Authorized responders supply updates; a coordinator confirms resolution. The complete event history remains available for review.

Terminology

Multi-agency emergency coordination: multiple services share incident information and coordinate roles. Mutual aid: another agency provides agreed support when capacity or specialist resources are needed. Volunteering: trained, verified people support suitable tasks under an authorized coordinator; it is a separate later-phase capability.

MVP demonstration: three simulated agency accounts (fire, ambulance and police), one busy agency, one accepting agency and one support request. No claim of live official integration. Real alerts require agency agreements, verified contacts and a tested escalation process.

06 / STACK

Recommended MVP stack and why

Choose a small Google Cloud stack that matches the team and its React and Node.js experience. These are proposed choices; confirm current service setup, quotas, costs and access before implementation.

Layer	Proposed choice	Reason
Frontend	React + TypeScript + Vite	Familiar stack for report intake, live map and agency/coordinator dashboards.
UI	Tailwind + shadcn/ui	Consistent forms, dialogs, incident cards and status controls.
Backend	Node.js + Express on Cloud Run	Server-owned validation, agency matching, assignment and escalation rules; familiar implementation path.
Database	Firestore	Incident records and real-time status subscriptions. Separate reports, incidents, agency profiles, assignments and events.
Authentication	Firebase Authentication	Agency and coordinator identities. Backend verifies tokens and enforces roles; frontend checks alone are insufficient.
AI	Gemini API	Multilingual extraction and clarification drafts, validated against a schema; missing facts stay unknown.
Map	Google Maps	Display incident and agency locations. Confirm billing, API restrictions and map usage requirements.
Notifications	Firebase Cloud Messaging + inbox	Push to enrolled devices plus a persistent dashboard record. Delivery failure must remain visible.
Hosting	Firebase Hosting + Cloud Run	Static frontend and API deployment on Google Cloud, aligned with the competition requirements.

Supporting backend design

Use geohash/radius candidate lookup followed by exact distance, coverage and capability checks. Store availability with a last-updated time. Run acknowledgement timeouts through a durable scheduled worker or task mechanism; do not rely on in-memory timers. Make retries idempotent and assignments transactional.

Start with simulated notifications and agencies. Nearby public alerts, volunteer onboarding and historical analytics follow after the core coordination flow works. There are no deployed capabilities or measured response-time improvements yet.

07 / EDGE CASE

Fire damages power and communication networks

Failure scenario

A fire damages building wiring and power infrastructure. Wi-Fi becomes unavailable; mobile coverage is weak, congested or absent. The reporter may also have low battery or a damaged device. In this situation, a cloud-based SOS system cannot assume that a report reaches the backend or that an agency receives an alert.

Expected behavior by connectivity state

Condition	Proposed behavior and limit
Internet available	Send the SOS and show its delivery state. Backend receipt, agency acknowledgement and responder assignment remain separate statuses.
Internet unavailable; cellular service available	Offer a call/SMS fallback where supported and configured. A call or SMS attempt is not proof of delivery, acceptance or dispatch. No agency integration is assumed.
No internet or cellular service	Allow local report capture if the device and app remain usable. Clearly show: SOS pending - not delivered. The web app cannot remotely alert agencies without a working communication channel.
Connection returns	Retry pending reports with a stable report ID to prevent duplicates. Preserve original creation time and label delayed delivery. Ask for a current situation update; old reports must not appear as fresh observations.
Cached information only	Display downloaded building information and incident history with last-sync time. Cached maps or statuses are not a live picture and do not establish a safe route.
Power or device failure	Software cannot operate on an unusable device. Building alarms and established emergency procedures must work independently of this app.

MVP mitigation

Propose a PWA with a service worker for the basic app shell and IndexedDB for locally queued text reports. Provide explicit retry on reopening and a manual send action; do not depend solely on background sync. Offline reporting uses structured manual fields because cloud Gemini processing is unavailable. Local storage is best-effort and can be cleared or lost.

Future resilience roadmap - outside the software MVP

Explore battery-backed physical SOS buttons, independent building alarms and a local gateway with an alternative communication channel, such as dedicated radio where appropriate. This requires building partners, qualified hardware specialists, site surveys and operational testing. A battery or gateway alone does not guarantee communication beyond the building.

08 / BUILD PLAN

A focused prototype and validation plan

Fresh project for this competition

The competition FAQ excludes pre-existing projects and submissions based on them. Build a fresh repository and implementation during the hackathon timeline. Prior rescue experience informs the team; ResQ or RescueX code must not be repackaged. Organizer clarification is needed for any uncertain reuse. [3]

Component	Proposed implementation
Frontend	React + TypeScript + Vite: text SOS intake, live map, incident cards and role-based dashboards.
AI layer	Gemini: schema-constrained extraction, summary and clarification drafts; server-side validation.
Backend and records	Node.js / Express on Cloud Run with Firestore for geolocated reports, incidents, agencies, assignments and audit events.
Identity and hosting	Firebase Authentication with server-enforced roles; Firebase Hosting or Cloud Run.
Demo dataset	Synthetic reports only, including Roman Urdu, English, ambiguity, duplicates and conflicting updates.

Schedule: 10-18 October

10-11: Finalize team, problem and synthetic scenarios. 12-13: Intake, extraction and editable cards. 14-15: Verified map, simulated agency alerts, acknowledgement and timeline. 16: Failure and access checks. 17: Cloud deployment, offline edge-case checks, video and proposal deck. 18: Final verification and submission.

Success criteria - targets, not measured results

Evaluate on 30-50 labeled synthetic reports. Target at least 90% correct extraction of explicitly stated fields and zero invented locations or counts in the test set. Compare manual processing with AI-assisted review; aim for a 30% reduction in median review time without reducing accuracy. Track duplicate false positives, correction rate and failed submissions.

Before a real pilot, interview at least one building manager and one emergency professional. Validate the reporting workflow and operational ownership; do not infer adoption or lives saved from a hackathon demo.

Phase 1: SOS intake, backend, verified live map, coordinator workflow and simulated multi-agency acknowledgement. Phase 2: opt-in nearby alerts and searchable location history. Phase 3: repeat-incident analytics, agreed agency integrations and verified volunteer support. Voice, camera detection, indoor navigation and autonomous dispatch remain outside the initial prototype.

09 / SUBMISSION & SOURCES

What we would present

A three-minute demonstration

Show one simulated building incident: submit a Roman Urdu report, inspect extracted fields, answer a clarification, review a possible duplicate, verify its map marker, notify simulated agencies, show one busy agency, confirm a lead/support assignment and record an update. Include an ambiguous report and an AI failure so judges can see the uncertainty and manual fallback behavior.

Competition fit

Proposed theme: Sustainability & Social Impact. Required delivery includes a working deployed prototype, public GitHub repository, short demo video and explanatory deck. The rules require Google AI models or specified agentic tools and Google Cloud deployment using Cloud Run or Firebase. Team formation closes 11 October; prototype submission closes 18 October 2026. [3-5]

Team decisions to confirm

Confirm team eligibility and membership; choose a single building scenario; agree on frontend/backend ownership; confirm cloud access and budget; decide who performs coordinator interviews. A named partner or emergency-service integration should only appear after it is actually agreed.

Sources and evidence notes

[1] Associated Press, 18 January 2026. "Death toll rises to 6 in fire at a shopping plaza in the Pakistani city of Karachi." Early reporting, used for incident context only; the headline is not a final casualty count.
Read AP report

[2] Reuters, 19 January 2026. "Dozens missing after massive Karachi mall fire, 21 killed." Used for reported smoke and ventilation barriers, not final casualty totals.
Read Reuters report

[3] AI Builder Cup FAQ: fresh-project eligibility, team and submission rules.
https://aibuildercup.com/Faqs.html

[4] AI Builder Cup challenges and requirements: themes, Google AI and deployment requirements.
https://aibuildercup.com/themes.html

[5] AI Builder Cup overview: official timeline.
https://aibuildercup.com/

Sources checked 10 October 2026. Product capabilities, differentiation and targets in this document are proposed, not implemented or independently validated. The Gul Plaza incident is the motivation, not evidence that this particular tool would have changed its outcome.